

Cooperative **Multiplex GNN** for High-Grade Glioma **Survival Prediction** from Preoperative **Multi-Modal Radiomics-Based Brain Networks**

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Cooperative Multiplex GNN for High-Grade Glioma Survival Prediction from Preoperative Multi-Modal Radiomics-Based Brain Networks

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Abstract

Accurately and preoperatively predicting survival for high-grade gliomas (HGGs) is important for optimizing treatment strategies. Increasing evidence suggests that brain structural and functional connectivity networks derived from advanced magnetic resonance imaging (MRI) are promising predictors for HGG survival. However, advanced MRIs (e.g., diffusion MRI and functional MRI) are generally clinically inaccessible for HGG patients before initiating therapy. To compensate for lack of advanced MRI modalities in brain network studies, in this paper we evaluate the feasibility and performance of predicting HGG survival using exclusively preoperative multi-modal basic structural MRI (sMRI, e.g., T1- and T2-weighted MRI) based brain regional radiomics similarity networks (R2SNs). To this end, we propose a new cooperative multiplex graph neural network (GNN) based multi-modal R2SN integration framework for preoperative HGG survival prediction. First, multi-modal R2SNs are represented by a multiplex network, where each modality-specific R2SN forms one multiplex layer and nodes (i.e., brain regions of interest (ROIs)) are coupled to their replicas across multiplex layers. This facilitates flexible inter-ROI communications both within and between R2SNs. Second, a cooperative GNN is applied to capture intra-modal node feature propagations within each multiplex layer, followed by attention mechanisms used to capture inter-modal node feature interactions across multiplex layers. Finally, a tailored tumor-aware graph pooling is developed to assemble features from the tumor-intersecting ROIs for survival prediction. Extensive experiments on a collected HGG database with three basic sMRI modalities demonstrate the superiority of our method over state-of-the-art baselines in survival stratification. The code is available at <https://github.com/ZiLaoTou/TCM-GNN>.

- Introduction
- Method
- Experimental Results
- Discussions
- Conclusion

- **High-Grade Glioma (HGG)**

- HGG는 WHO 분류 기준 Grade III-IV에 해당하는 악성 뇌종양으로, 전체 glioma의 60-75%를 차지하며 예후가 극히 불량한 종양군
- 따라서 수술 전 HGG 생존을 예측하는 것은 위험 수준이 다른 환자에 대한 맞춤형 치료 결정을 내리는 데 중요

- **sMRI (structural MRI)**

- 뇌의 구조적 정보를 촬영 {T1c, T2, Flare} 는 각자 강조하는 부분이 다름.
- 초기 연구들은 preoperative sMRI 이미지 자체나 거기서 뽑은 radiomics feature를 CNN, random forest 등에 입력해 생존을 예측
 - Multi-modal sMRI를 함께 쓰면 정확도가 상승
 - 이 방법들은 전체 뇌나 개별 ROI를 독립적으로 봤을 뿐, ROI간 관계를 무시

- **Regional Radiomics Similarity Networks (R2SN)**

- dMRI 기반 SCN과 fMRI 기반 FCN을 활용한 연구가 있었으나, 임상에서는 치료전 dMRI나 fMRI를 찍지 않는 것이 대부분 (specialized equipment, cost, and clinical expertise)
- 그렇다면 누구나 갖고 있는 기본 sMRI(T1c, T2, FLAIR)로 brain network를 만들 수 있을까?
- R2SN은 기본 sMRI에서 ROI별 radiomics feature를 추출하고, ROI 쌍 간의 Pearson's correlation으로 edge를 정의해 구성하는 network
- dMRI·fMRI 없이도 뇌의 anatomical connectome을 의미 있게 표현할 수 있으며, Alzheimer's disease 진단에서 이미 유효성이 검증된 바 있음

- **Multiplex Network**

- Multi R2SN을 multiplex network로 표현하여 inter-/intra- modality간 통신을 구조적으로 허용
- cooperative multiplex graph neural network (CM-GNN)으로 intra-modal node feature를 동적으로 propagate하고, attention으로 inter-modal feature를 집계
- 종양과 겹치는 ROI들만 pooling하는 Tumor-Aware Graph Pooling → MLP로 생존 이진 분류

Multi-Modal R2SN Construction and Representation by a Multiplex Network

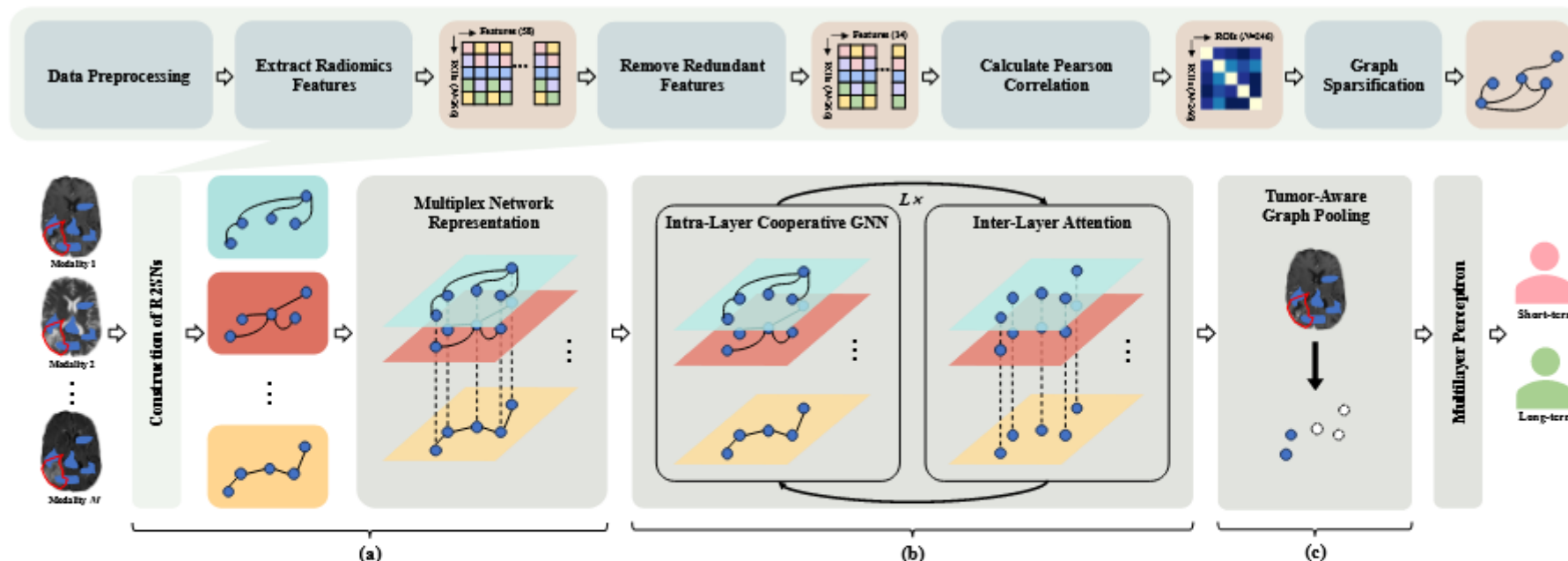
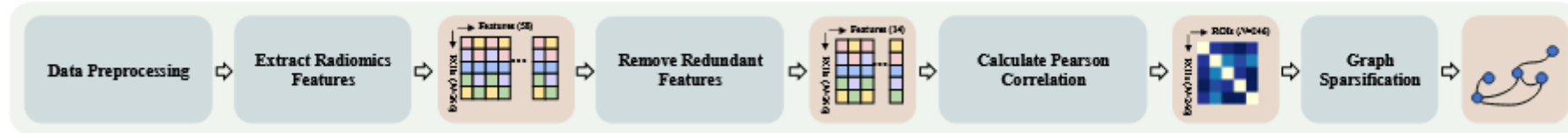


Fig. 1: Illustration of the proposed CM-GNN based multi-modal R2SN fusion for brain tumor survival prediction. (a) M R2SNs are constructed from preoperative M basic sMRI modalities separately, and then modeled by a multiplex network. (b) The node embeddings are extracted by an L -layer CM-GNN, in which cooperative GNN based intra-modal node feature propagations and attention based inter-modal node feature interactions are alternatively implemented. (c) Prior to feeding into an MLP classifier for survival prediction, the obtained node embeddings of only those ROIs that intersect with tumor areas are pooled by Tumor-Aware Graph Pooling. Brain area enclosed by the red line represents the tumor, and the blue areas represent ROIs.

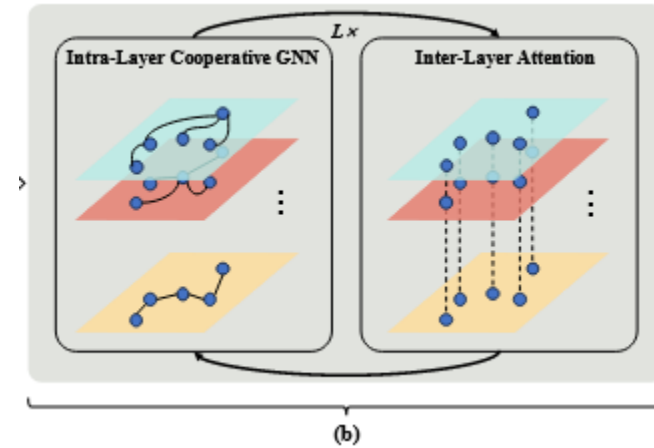
• Multi-Modal R2SN Construction and Representation by a Multiplex Network



- $1 \leq m \leq M$ (M 은 modal의 수)
- N : Brainnetome Atlas (246 ROIs)
- d : intensity- and texture based radiomics features (18+40=58)
- radiomics feature matrix: $F_m \in R^{N \times d_m}$
- R2SN graph: $G_m \triangleq (V, A_m, F_m)$
 - V : node set
 - A_m : adjacency matrix ROI 쌍 간 radiomics similarity (30% largest edge weight)
 - F_m : node feature matrix

- **Multi-Modal R2SN Fusion by CM-GNN**

- 각 레이어 별로 2단계 연산
- 각 modality layer 내부에서 ROI 간 정보 교환 (Intra-modal)
 - 일반 GNN과 다르게 CM-GNN은 모든 이웃끼리 정보교환을 하지 않음
 - GNN_{action} : { listen and broadcast, listen, broadcast, isolate }의 상태를 배정하여 어떤 이웃과 정보를 교환할지 결정
 - $GNN_{environment}$: 살아남은 edge로만 정보를 교환해서 각 ROI의 representation을 업데이트
- M modality 간에 같은 ROI끼리 정보 교환 (Inter-modal)
 - Attention으로 modality 중요도 판단 후 통합
 - 최종 출력 $X^{(l)} = [x_1^{(l)}, x_2^{(l)}, \dots, x_N^{(l)}] \in R^{N \times d^{(l)}}$



- **Node Feature Aggregation by Tumor-Aware Graph Pooling**

- Tumor가 있는 부분의 ROIs만 Avg. pooling

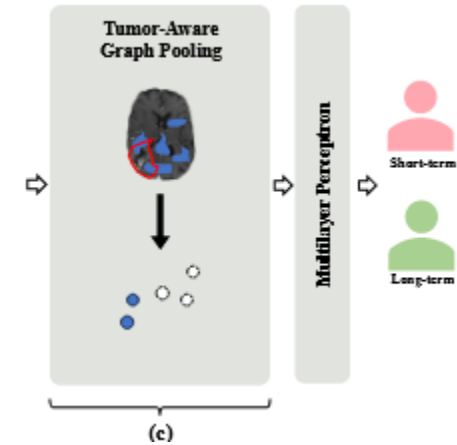
- $o = \frac{1}{||t||_1} (X^{(L)})^T t$ where $t \in R^N$ is indicator.

- **Loss Function of HGG Survival Prediction**

- Short term(≤ 1 year) vs Long term (> 1 year)

- $y_{pred} = \sigma(MLP(o))$

- $L = -y_{true} \cdot \log(y_{pred}) - (1 - y_{true}) \cdot \log(1 - y_{pred})$



- Datasets and Preprocessing**

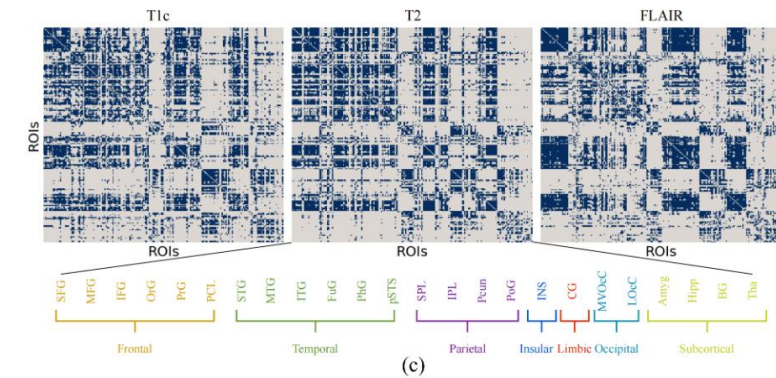
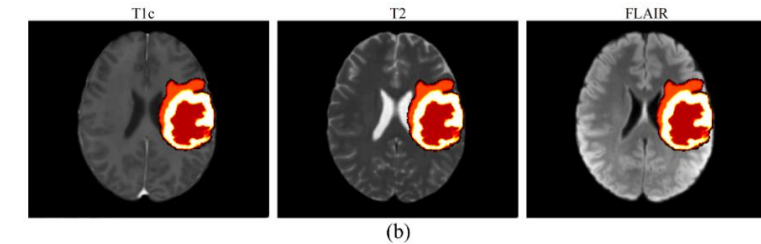
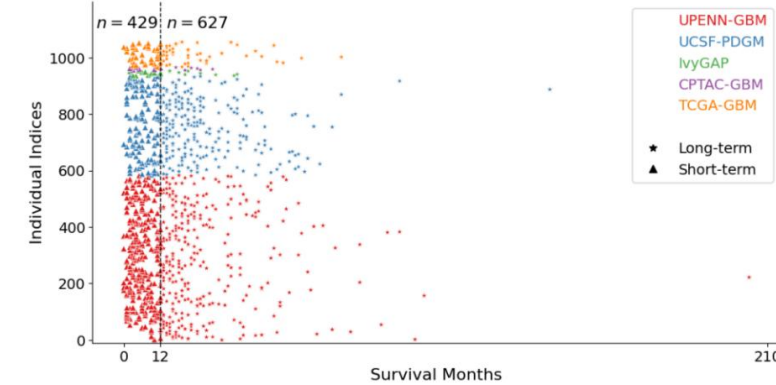
TABLE I: Demographic information of the HGG patients used in this study.

Cohort	Sample Size	Male/Female	Age (mean $\pm$ std)
UPENN-GBM	584	350/234	63.48 $\pm$ 11.89
UCSF-PDGM	351	213/138	58.36 $\pm$ 14.10
TCGA-GBM	89	58/31	59.81 $\pm$ 12.24
IvyGAP	20	9/11	58.70 $\pm$ 9.92
CPTAC-GBM	12	8/4	63.36 $\pm$ 11.23
All	1056	638/418	61.38 $\pm$ 12.87

- **Datasets and Preprocessing**

- T1c, T2, FLAIR images + tumor seg. annot.

1. Brain extraction and co-registration
2. Standard preprocessing pipeline using the Advanced Normalization Tools (ANTs)
 1. noise reduction with a spatially adaptive filter
 2. initial intensity normalization
 3. N4 deviation field correction (T1c, T2)
 4. a second intensity normalization (T1c, T2)
 5. registration to the MNI-152 1mm³ template by the SyN algorithm.



- **Implementation Details**
- Data-splitting bias 제거를 위해 10-times 5-fold cross validation
- Metric: AUCROC, ACC, precision, recall, F1
- Env
 - Ubuntu 20.04
 - RTX 4090 2개
 - python 3.10
 - PyTorch 2.0.1

GNN	
L	2
$d^{(1)} = d^{(2)}$	128
Attn. heads	4
Activation function	GELU
MLP	
Dim, L	64, 2
Drop out	0.1
Activation function	GELU
Training	
Batch size	48
Epoch	100
LR	$1e^{-4}$
Optimizer	AdamW
weight decay	$1e^{-5}$
Scheduler	cosine annealing

TABLE II: Classification results by different methods for HGG survival prediction; mean $\pm$ std.

Method	Modality			AUC	Accuracy	Precision	Recall	F1
	T1c	T2	FLAIR					
Radiomics-SF		Anyone		61.79 $\pm$ 3.61	59.87 $\pm$ 3.46	67.53 $\pm$ 4.33	62.78 $\pm$ 4.98	64.88 $\pm$ 3.98
Radiomics-CF	✓			60.64 $\pm$ 2.95	55.22 $\pm$ 2.91	67.46 $\pm$ 4.12	47.83 $\pm$ 4.58	55.55 $\pm$ 5.32
		✓		61.26 $\pm$ 3.49	57.70 $\pm$ 2.97	68.54 $\pm$ 3.78	53.48 $\pm$ 4.36	59.93 $\pm$ 3.03
			✓	59.43 $\pm$ 3.02	54.93 $\pm$ 3.10	67.69 $\pm$ 3.82	46.39 $\pm$ 4.52	54.89 $\pm$ 3.46
Radiomics	✓			62.02 $\pm$ 3.23	59.93 $\pm$ 3.02	66.21 $\pm$ 4.26	66.82 $\pm$ 4.75	66.35 $\pm$ 3.07
		✓		62.64 $\pm$ 3.89	60.31 $\pm$ 2.68	66.23 $\pm$ 3.17	67.88 $\pm$ 4.32	66.93 $\pm$ 2.53
			✓	62.43 $\pm$ 3.07	60.64 $\pm$ 2.54	66.78 $\pm$ 3.95	67.62 $\pm$ 4.13	67.03 $\pm$ 2.29
SVC	✓			60.13 $\pm$ 3.34	59.14 $\pm$ 2.66	65.52 $\pm$ 3.50	66.11 $\pm$ 4.43	65.68 $\pm$ 2.70
		✓		61.76 $\pm$ 3.12	60.21 $\pm$ 2.94	66.24 $\pm$ 3.44	67.48 $\pm$ 4.09	66.75 $\pm$ 2.70
			✓	61.28 $\pm$ 2.84	59.81 $\pm$ 2.58	65.85 $\pm$ 3.79	67.47 $\pm$ 4.08	66.52 $\pm$ 2.63
GCN	✓			58.39 $\pm$ 4.00	57.27 $\pm$ 3.48	63.97 $\pm$ 4.05	63.46 $\pm$ 4.03	63.71 $\pm$ 4.04
		✓		60.76 $\pm$ 3.75	59.53 $\pm$ 3.08	65.90 $\pm$ 3.61	65.38 $\pm$ 3.60	65.64 $\pm$ 3.60
			✓	56.77 $\pm$ 3.37	56.16 $\pm$ 3.54	63.05 $\pm$ 3.77	62.55 $\pm$ 3.75	62.80 $\pm$ 3.76
GAT	✓			58.76 $\pm$ 3.67	57.39 $\pm$ 3.20	64.08 $\pm$ 3.75	63.57 $\pm$ 3.74	63.83 $\pm$ 3.74
		✓		62.26 $\pm$ 3.79	60.49 $\pm$ 3.24	66.73 $\pm$ 3.62	66.20 $\pm$ 3.62	66.46 $\pm$ 3.62
			✓	58.06 $\pm$ 2.87	57.41 $\pm$ 3.03	64.12 $\pm$ 3.38	63.61 $\pm$ 3.37	63.86 $\pm$ 3.37
BrainGNN	✓			61.84 $\pm$ 3.45	59.83 $\pm$ 3.39	66.15 $\pm$ 3.88	65.62 $\pm$ 3.87	65.89 $\pm$ 3.88
		✓		66.59 $\pm$ 3.74	64.01 $\pm$ 3.37	69.76 $\pm$ 3.36	69.20 $\pm$ 3.35	69.48 $\pm$ 3.36
			✓	64.35 $\pm$ 4.11	61.86 $\pm$ 3.64	67.84 $\pm$ 4.19	67.30 $\pm$ 4.18	67.57 $\pm$ 4.19
BNT	✓			63.57 $\pm$ 3.56	61.46 $\pm$ 3.10	67.57 $\pm$ 3.35	67.03 $\pm$ 3.35	67.30 $\pm$ 3.35
		✓		65.97 $\pm$ 3.02	62.57 $\pm$ 2.91	68.53 $\pm$ 3.07	67.98 $\pm$ 3.06	68.26 $\pm$ 3.06
			✓	63.99 $\pm$ 3.58	61.63 $\pm$ 3.28	67.73 $\pm$ 3.33	67.19 $\pm$ 3.32	67.46 $\pm$ 3.33
CoGNN	✓			64.74 $\pm$ 3.58	62.16 $\pm$ 3.17	68.15 $\pm$ 3.47	67.61 $\pm$ 3.46	67.88 $\pm$ 3.47
		✓		65.52 $\pm$ 3.48	62.08 $\pm$ 3.06	68.09 $\pm$ 3.38	67.55 $\pm$ 3.38	67.82 $\pm$ 3.38
			✓	64.54 $\pm$ 2.97	61.63 $\pm$ 2.89	67.70 $\pm$ 3.40	67.16 $\pm$ 3.40	67.42 $\pm$ 3.40
SVC-MM	✓	✓	✓	65.32 $\pm$ 3.37	62.48 $\pm$ 2.91	67.80 $\pm$ 3.72	70.44 $\pm$ 4.62	68.95 $\pm$ 2.74
FastGTN	✓	✓	✓	66.48 $\pm$ 3.21	62.86 $\pm$ 2.92	68.76 $\pm$ 3.14	68.22 $\pm$ 3.13	68.49 $\pm$ 3.13
RH-BrainFS	✓	✓	✓	62.92 $\pm$ 2.48	61.04 $\pm$ 3.06	67.17 $\pm$ 3.72	66.63 $\pm$ 3.71	66.90 $\pm$ 3.72
Cross-GNN	✓	✓	✓	66.81 $\pm$ 3.27	63.26 $\pm$ 3.06	69.09 $\pm$ 3.36	68.54 $\pm$ 3.35	68.81 $\pm$ 3.36
Multiplex-GNN	✓	✓	✓	68.11 $\pm$ 3.58	64.34 $\pm$ 3.03	70.06 $\pm$ 2.90	69.50 $\pm$ 2.89	69.77 $\pm$ 2.90
Ours	✓	✓	✓	72.21 $\pm$ 2.72	67.77 $\pm$ 3.24	72.94 $\pm$ 3.25	72.36 $\pm$ 3.24	72.65 $\pm$ 3.25

- Ablation study**

TABLE III: Ablation results for the GNN_{action} , inter-layer attention mechanism, and Tumor-Aware Graph Pooling in our method.

Method Variant	AUC	Accuracy	Precision	Recall	F1
w/o GNN_{action}	71.55 ± 2.58	66.87 ± 2.66	72.19 ± 2.75	71.61 ± 2.75	71.90 ± 2.75
w/o Inter-layer Attention	71.58 ± 2.85	67.39 ± 2.67	72.62 ± 2.82	72.04 ± 2.82	72.33 ± 2.82
w/o Tumor-Aware Graph Pooling	69.36 ± 2.75	65.21 ± 3.00	70.76 ± 3.18	70.19 ± 3.18	70.47 ± 3.18
Full Method (Ours)	72.21 ± 2.72	67.77 ± 3.24	72.94 ± 3.25	72.36 ± 3.24	72.65 ± 3.25

TABLE IV: Classification results by our method using single-modal, two-modal, and three-modal R2SNs, respectively.

Modality			AUC	Accuracy	Precision	Recall	F1
T1c	T2	Flair					
✓			61.78 ± 4.34	60.55 ± 3.50	66.74 ± 4.09	66.21 ± 4.08	66.48 ± 4.08
	✓		67.53 ± 3.50	65.04 ± 3.66	70.58 ± 3.94	70.02 ± 3.93	70.30 ± 3.93
		✓	63.58 ± 3.26	61.97 ± 3.08	68.00 ± 3.32	67.46 ± 3.31	67.73 ± 3.32
✓	✓		71.56 ± 3.04	67.27 ± 3.13	72.54 ± 3.00	71.96 ± 2.99	72.25 ± 3.00
✓		✓	68.59 ± 3.34	65.34 ± 3.09	70.87 ± 3.21	70.31 ± 3.20	70.59 ± 3.20
	✓	✓	71.37 ± 2.59	67.37 ± 2.57	72.62 ± 2.60	72.04 ± 2.59	72.33 ± 2.59
✓	✓	✓	72.21 ± 2.72	67.77 ± 3.24	72.94 ± 3.25	72.36 ± 3.24	72.65 ± 3.25

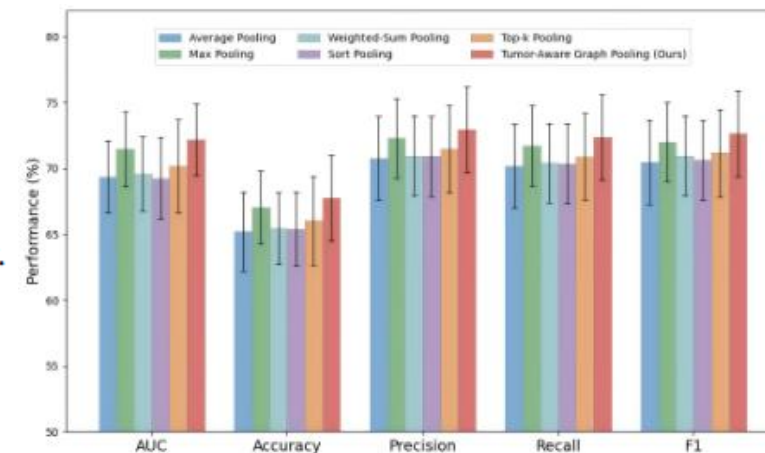


Fig. 3: Classification results for various pooling strategies used in our method.

- **Leave-One-Cohort-Out (LOCO)**

- UPENN-GBM – 584 patients
- CPTAC-GBM – 12 patients

- **Sensitivity Analysis of the Sparsification Threshold**

TABLE V: Classification performance by different methods based on LOCO.

Test Cohort	Method	AUC	Accuracy	Precision	Recall	F1
IvyGAP	Cross-GNN	58.56 $\pm$ 7.60	56.11 $\pm$ 7.37	56.79 $\pm$ 8.19	51.11 $\pm$ 7.37	53.80 $\pm$ 7.76
	Multiplex-GNN	57.90 $\pm$ 6.17	58.00 $\pm$ 4.58	58.89 $\pm$ 5.09	53.00 $\pm$ 4.58	55.79 $\pm$ 4.83
	Ours	64.80 $\pm$ 3.31	65.00 $\pm$ 6.32	66.67 $\pm$ 7.03	60.00 $\pm$ 6.32	63.16 $\pm$ 6.66
TCGA-GBM	Cross-GNN	66.49 $\pm$ 3.61	62.92 $\pm$ 2.46	67.35 $\pm$ 2.23	66.00 $\pm$ 2.19	66.67 $\pm$ 2.21
	Multiplex-GNN	64.14 $\pm$ 4.74	60.45 $\pm$ 3.95	65.10 $\pm$ 3.59	63.80 $\pm$ 3.52	64.45 $\pm$ 3.55
	Ours	71.27 $\pm$ 1.72	64.94 $\pm$ 2.92	69.19 $\pm$ 2.65	67.80 $\pm$ 2.60	68.49 $\pm$ 2.63
UCSF-PDGM	Cross-GNN	61.82 $\pm$ 2.94	62.79 $\pm$ 2.60	72.66 $\pm$ 1.92	72.35 $\pm$ 1.91	72.50 $\pm$ 1.92
	Multiplex-GNN	67.90 $\pm$ 3.24	67.07 $\pm$ 2.22	75.82 $\pm$ 1.65	75.50 $\pm$ 1.64	75.66 $\pm$ 1.64
	Ours	72.55 $\pm$ 1.20	68.84 $\pm$ 1.19	77.13 $\pm$ 0.88	76.81 $\pm$ 0.88	76.97 $\pm$ 0.88

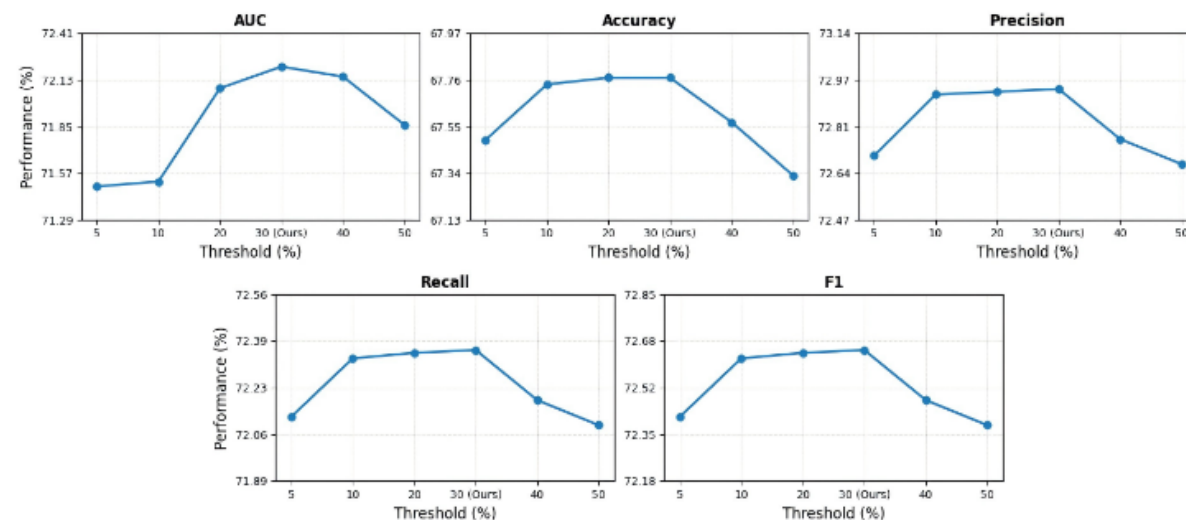


Fig. 4: Classification results for various graph sparsification thresholds applied in our method.

• Differences in Multiplex Network Topology Between HGG Short- and Long-Term Survival Groups

- short-term 환자가 long-term 환자보다 대부분의 ROI에서 multiplex degree가 더 높음.
- 가장 유의미한 Top-5 ROI는 인지, 감정, 언어와 관련

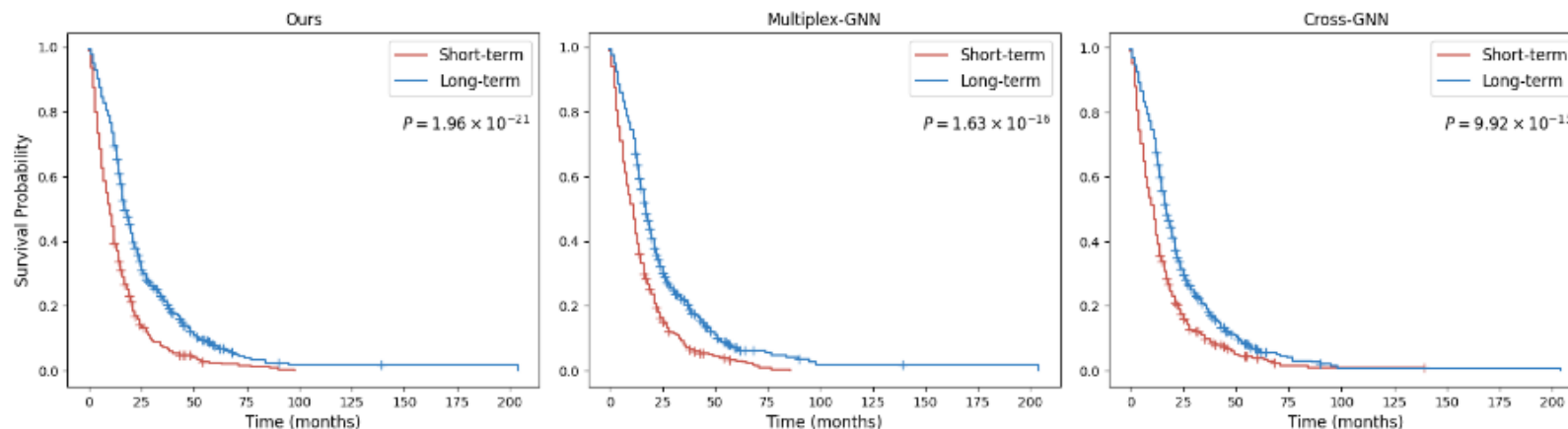


Fig. 6: The Kaplan-Meier survival curves and the log-rank test based P -values between the HGG short- and long-term survival groups by our method, Multiplex-GNN, and Cross-GNN.

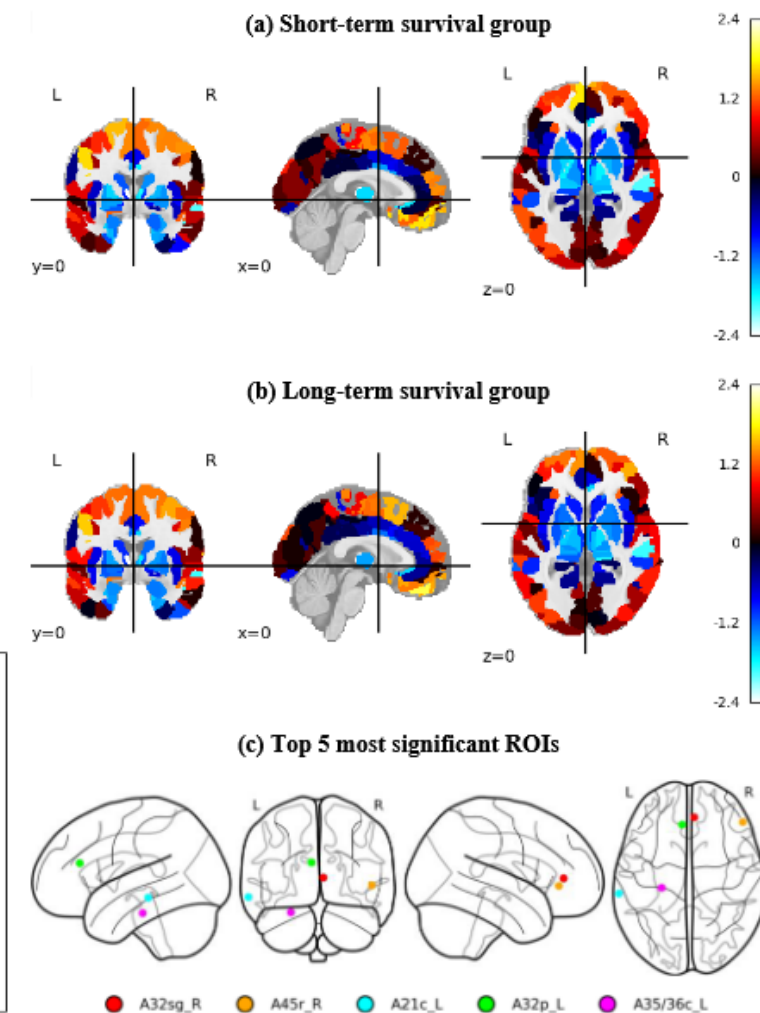


Fig. 5: (a)&(b) The mean multiplex degrees of nodes displayed as color-coded maps for the HGG short- and long-term survival groups. (c) The top 5 most significant ROIs for HGG survival.

- **Limitations**

- 3개 이상의 ROIs 관계는 고려하지 않음 (pair-wise)
- GNN이라 Radiomics feature interpretability 부족
- Tumor-Aware Pooling이 ED, ET, NCR/NET을 동등하게 처리
- Cohort 간 data heterogeneity 미반영
- Missing modality